NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

CCDS25-0096

FORMALIZING THE ISOLATION LEMMA IN ISABELLE/HOL

Jiang Yifei

Supervisor: Assistant Professor Tan Yong Kiam

COLLEGE OF COMPUTING AND DATA SCIENCE

Submitted in Partial Fulfilment of the Requirements
for the Degree of Bachelor of Computing in Computer Science
at Nanyang Technological University

2026

ABSTRACT

Randomized algorithms are a cornerstone of modern theoretical computer science [1], yet their proofs often rely on informal probabilistic intuition that obscures critical assumptions about independence and measurability. This report presents a formal verification of the **Isolation Lemma**, a key tool in randomized algorithms that isolates a unique minimum-weight solution by randomly assigning weights to combinatorial problems such as the **Perfect Matching** problem [2], [3]. By carrying out this formalization in the **Isabelle/HOL** [4], [5] interactive theorem prover, we make all probabilistic assumptions explicit and machine-checkable, thereby strengthening the rigor of the underlying algorithmic reasoning [6].

We begin by formally specifying the Isolation Lemma, including the underlying product probability space for random weight assignments, the precise statement of the lemma, and the combinatorial constructions required to reason about minimizers over finite sets. Building on this foundation, we develop a complete machine-checked proof by synthesizing and reconciling arguments scattered across multiple informal and folklore-style presentations of the lemma and resolving implicit assumptions present in their informal treatments. Finally, we apply the formalized Isolation Lemma to verify a randomized parallel algorithm for the **Perfect Matching** problem, demonstrating how classical randomized algorithmic techniques can be systematically integrated into a rigorous formal verification framework.

ACKNOWLEDGMENTS

I would like to express my gratitude to Assitant Professor Tan Yong Kiam for his guidance and support throughout this project.

TODO: Acknowledgements

TABLE OF CONTENTS

Abstract ..... **ii**

Acknowledgements ..... **iii**

1 Introduction **1**

1.1 Background 2

1.1.1 Randomized Algorithms in TCS 2

1.1.2 Formalization and Theorem Proving 3

1.2 Project Objectives 4

1.3 Scope of Work 5

1.4 Report Outline 5

2 Preliminaries **7**

2.1 Interactive Theorem Proving 7

2.2 Isabelle/HOL System 8

2.3 The Isolation Lemma 8

2.4 Parallel Algorithm for Perfect Matching 11

2.4.1 The Perfect Matching Problem 11

2.4.2 Challenges in Parallel and Algebraic Approaches 12

2.4.3 Random Weighting and the Isolation Principle 13

2.4.4 Role of the Isolation Lemma in Perfect Matching Algorithms 14

3 Formal Modeling of the Isolation Lemma **15**

3.1 Modeling Weights and Universe 16

3.2 Defining Minimizers and Uniqueness 18

3.3 The Alpha Construction 20

3.4 Probability Space Construction 21

4 Proof Construction **24**

4.1 Existence of Minimizers 25

4.2 Reformulating the Non-Uniqueness 26

4.3 The Alpha Construction 28

4.3.1 Simplifying the Definition of $\alpha(x)$ 28

4.3.2 Independence of $\alpha(x)$ from Individual Weights 30

4.3.3 Collision characterization via $\alpha(x)$ 31

4.4 Probabilistic Analysis and Final Bound 33
4.5 Chapter Summary 37
5 Perfect Matching Problem **38**
5.1 Introduction to Randomized Parallel Matching 38
5.1.1 Decision versus Construction 38
5.1.2 The Symmetry Barrier and Coordination 39
5.1.3 Isolation as a Solution 39
5.2 Algebraic Modeling: The Tutte Matrix 40
5.2.1 Matrix Construction in Isabelle/HOL 40
5.2.2 Determinant and Perfect Matchings 40
5.3 The Weight-to-Multiplicity Bridge 41
5.3.1 The Non-Cancellation Principle 41
5.3.2 Formalization via Multiplicity 42
5.4 Parallel Edge Extraction Logic 43
5.4.1 Minors and Subgraph Matchings 43
5.4.2 The Parallel Decision Predicate 43
5.5 Probabilistic Correctness via Isolation Lemma 44
5.5.1 Mapping Graphs to Sets 45
5.5.2 Final Probability Bound 45
5.6 Formalization Challenges in Isabelle/HOL 46
5.6.1 Bridging Type Systems: Int versus Nat 46
5.6.2 Library Integration and Automation 47
5.7 Chapter Summary 47
6 Conclusion and Future Work **49**
6.1 Summary of Research Achievements 49
6.2 Methodological Lessons and Reflections 49
6.2.1 The Ontology of Randomness: From Intuition to PMF Objects . 50
6.2.2 Structural Bridging: The Utility of the SP Layer 50
6.2.3 Representation and Consistency: Permutations versus Sets 50
6.2.4 The Hidden Burden of Rigor: Type Safety and Measurability ... 50
6.3 Future Work and AFP Submission 51

References **52**

1 INTRODUCTION

Randomization has become one of the central paradigms of modern theoretical computer science (TCS) [1]. Over the past few decades, randomized methods have shaped the design and analysis of algorithms, yielding constructions that are simpler, faster, or in some settings seemingly out of reach for known deterministic techniques.

At the heart of randomized algorithms lies a collection of probabilistic lemmas that explain why randomness enforces useful structural properties – for instance, why a random perturbation can isolate a unique optimum among many candidates. These lemmas are routinely invoked across areas such as parallel computation, combinatorial optimization and matching.

However, the proofs of such lemmas are often developed as pen-and-paper arguments, or circulate as folklore-style proofs that compress multiple layers of reasoning into a few lines. In these presentations, assumptions about probability spaces, measurability, independence, and minimization are commonly treated implicitly, and intermediate steps are omitted.

This proof style can be fragile: the intricate details sometimes fail to survive close mathematical scrutiny, especially when probabilistic reasoning interacts with nontrivial algorithmic constructions. Concrete instances of this phenomenon have appeared in recent work, where closer inspection or formal reconstruction revealed missing conditions or minor errors in widely cited arguments, both in cryptographic settings and in the analysis of advanced algorithmic techniques [7], [8].

Interactive theorem proving (ITP) provides a disciplined response to this challenge. By requiring definitions and proof steps to be stated explicitly and checked mechanically, ITP strengthens rigor, turns informal mathematics into machine-checkable artifacts, and supports large-scale collaboration on proofs. Recent developments demonstrate that such mechanization can scale to nontrivial randomized algorithm analyses, for example by formally verifying the CVM distinct-elements streaming algorithm in Isabelle/HOL using functional probabilistic invariants to derive concentration bounds [9]. Moreover, the resulting formal

developments can serve as reusable digital building blocks, contributing to the broader effort of digitalizing mathematics and enabling human–AI collaboration in formal reasoning.

This report contributes to the formal verification of probabilistic reasoning in TCS by formalizing the *Isolation Lemma* in Isabelle/HOL. The Isolation Lemma is a key tool that assigns random weights to enforce, with high probability, a unique minimum-weight solution structure; it is notably used in randomized algorithms for the **Perfect Matching** problem. Our formalization makes the underlying probability space and all dependencies explicit, and we further apply the lemma to support the verification of a randomized perfect matching algorithm.

1.1 Background

1.1.1 Randomized Algorithms in TCS

Randomized algorithms are a widely studied class of algorithms in theoretical computer science.

Unlike deterministic algorithms, randomized algorithms incorporate randomness to break symmetry (often referred to as the *symmetry barrier* in parallel complexity [2]), simplify theoretical constructions, and achieve improved expected performance. Such algorithms have been widely used in matching problems, parallel computing, approximation algorithms [10], and the construction of complex combinatorial structures.

In many classical results, the introduction of randomized weights serves as a general technique for enforcing certain "uniqueness" or "isolation" properties. For example, in several parallel algorithms and combinatorial optimization problems, algorithmic correctness relies on the fact that, after a random perturbation, a minimum structure becomes unique with high probability.

The Isolation Lemma, first introduced by Mulmuley, Vazirani, and Vazirani [2], is a canonical instance of this principle. Informally, this lemma states that when random

weights are assigned independently to elements of a finite universe, there is a high probability that a unique minimum-weight set exists within a given family of subsets. This lemma provides a theoretical foundation for numerous randomized algorithms and plays a critical role in algorithmic frameworks for problems such as Perfect Matching [3].

1.1.2 Formalization and Theorem Proving

In theoretical computer science, many fundamental results rely on intricate mathematical arguments and precise logical reasoning. As theoretical frameworks continue to develop, the scale and complexity of such proofs have grown substantially, making traditional human-based verification increasingly difficult [6].

Formalized mathematics and interactive theorem proving (ITP) offer an alternative approach to reasoning about mathematical theories. In this setting, definitions, theorems, and proofs are encoded within a formal logical system [4], and every inference step is checked rigorously by a computer. This provides a level of precision and reliability that is difficult to achieve through informal reasoning alone.

A key characteristic of interactive theorem proving is that all assumptions must be stated explicitly. These include structural properties such as the non-emptiness and finiteness of sets, the totality of functions, and in probabilistic settings, the measurability of events and constructions of probability spaces. As a consequence, formal proofs do not rely on the intuition of the reader, but on the soundness of the underlying logical framework.

Beyond individual correctness guarantees, ITP also contributes to the digitalization of mathematical proofs. Formal developments can be stored, reused, and extended, enabling large-scale collaboration and incremental verification across research communities. This shift supports new forms of human–computer and human–AI collaboration in mathematics and theoretical computer science.

In recent years, formalized methods have been successfully applied across many areas in computer science, including program verification, compiler correctness [11], and the formalization of foundational mathematical theories [12]. Within theoretical computer

science, formal verification has gradually expanded from classical mathematical results to core theorems in algorithm theory and computational complexity.

However, the formal verification of randomized algorithms poses additional challenges. Randomized algorithms combine complex combinatorial reasoning with probabilistic concepts such as independence, distributions, and expectations, all of which must be expressed through explicit constructions within a rigorous measure-theoretic framework [5]. Against this background, the formal verification of key probabilistic lemmas has emerged as an important research direction in interactive theorem proving.

The Isolation Lemma serves as a representative example of such results. It combines combinatorial minimization with probabilistic reasoning and thus provides an ideal case study for examining how randomized algorithmic arguments can be translated into fully formal, machine-checked proofs.

1.2 Project Objectives

The primary objective of this project is to formally model and prove the Isolation Lemma within the Isabelle/HOL theorem proving system. This includes establishing its probabilistic statement, associated probability bounds, and the precise conditions under which uniqueness holds.

To achieve this, the project explicitly formalizes the probabilistic and combinatorial structures underlying the lemma, including random weight assignments, independence of random variables, and minimization over finite sets.

Specifically, this project aims to:

- formalize random weight functions together with their corresponding product probability spaces, thereby making probabilistic assumptions explicit;
- provide a complete machine-checked proof of the Isolation Lemma, including an explicit upper bound on the probability of non-uniqueness and a precise characterization of failure conditions;

- apply the formalized lemma to the correctness analysis of randomized algorithms, supporting higher-level algorithmic reasoning.

As a concrete application, the Isolation Lemma is used to formally verify a key probabilistic argument in a randomized algorithm for the **Perfect Matching** problem.

1.3 Scope of Work

The scope of this project is focused on theoretical formalization and proof construction within the Isabelle/HOL framework. In particular, this report covers:

- the mathematical formulation of the Isolation Lemma and its formal expression;
- the probabilistic modeling of random weight assignments and their associated properties;
- the formalization of key combinatorial and probabilistic arguments appearing in the proof;
- a representative application of the lemma to the **Perfect Matching** problem.

This work does not address general derandomization techniques for randomized algorithms, nor does it consider implementation-level concerns or performance optimization. Moreover, the internal implementation details of the Isabelle/HOL system and its underlying logical foundations are beyond the scope of this report.

1.4 Report Outline

The remainder of this report is organized as follows.

Chapter 2 (Preliminaries) introduces the background material required for this work, including randomized algorithms, probabilistic reasoning, and basic concepts from interactive theorem proving. This chapter is limited to definitions and results that are not part of the author's original contributions, and serves to establish the theoretical context for the formal developments that follow.

Chapter 3 (Modeling) presents the formal modeling framework used throughout the report. It describes how the combinatorial and probabilistic structures underlying the Isolation Lemma are represented in Isabelle/HOL, including the construction of probability spaces and auxiliary definitions required for formal reasoning.

Chapter 4 (Proof) contains the core formalization of the Isolation Lemma. The development is organized using a PP/SP/FP structure – corresponding to pen-and-paper proofs (PP), structured proofs (SP), and fully formal proofs (FP) – which is carefully motivated and employed to connect informal probabilistic intuition, precise formal statements, and fully machine-checked formal verification.

Chapter 5 (Perfect Matching) applies the formalized Isolation Lemma to the verification of a randomized algorithm for the Perfect Matching problem. This chapter focuses on the high-level integration of the lemma into algorithmic reasoning.

Finally, Chapter 6 (Conclusion) summarizes the main contributions of the report and discusses possible directions for future work.

2 PRELIMINARIES

Before turning to the formal development of the Isolation Lemma, we briefly review the proof environment in which our formalization is carried out. This chapter introduces the basic ideas of interactive theorem proving and the Isabelle/HOL system, which together provide the logical and computational framework for the results developed in later chapters. The material presented here is purely preparatory and does not constitute original contributions.

2.1 INTERACTIVE THEOREM PROVING

Interactive theorem proving (ITP) is an approach to formal reasoning in which mathematical definitions, statements, and proofs are expressed within a precisely specified logical system and verified by a computer. Unlike fully automated theorem provers, interactive systems rely on human guidance to structure proofs, while delegating routine logical inference to the underlying proof engine.

A central characteristic of interactive theorem proving is that all logical assumptions must be made explicit. Properties that are often left implicit in pen-and-paper proofs – such as finiteness, non-emptiness, totality of functions, or measurability of events – must be stated precisely and verified within the formal system. As a result, proofs developed in an interactive theorem prover do not appeal to informal intuition, but instead derive correctness solely from the soundness of the underlying logic.

Interactive theorem proving has been successfully applied across a wide range of areas in computer science and mathematics, including program verification, compiler correctness, and the formalization of foundational mathematical theories. In the study of algoriths, ITP provides a principled framework for reasoning about complex algorithmic arguments, particularly when proofs involve subtle dependencies between combinatorial and logical assumptions.

From a methodological perspective, ITP also supports the digitalization of mathematical proof [6]. Formal developments can be stored as machine-checkable artifacts, reused across projects, and extended collaboratively. This enables

large-scale proof developments and supports emerging forms of human–computer and human–AI collaboration in formal reasoning.

2.2 Isabelle/HOL System

Isabelle is a generic interactive theorem proving framework that supports formal reasoning in a variety of logical systems. Among its most widely used instantiations is Isabelle/HOL, which is based on classical higher-order logic (HOL) [4]. In this setting, logical formulas are constructed using typed λ-calculus, and reasoning proceeds within a sound and expressive classical logic.

Isabelle/HOL provides a rich environment for formal mathematical development. Its logic supports higher-order functions, inductive definitions, and structured proofs, making it well suited for the formalization of mathematical arguments arising in theoretical computer science. The system is accompanied by an extensive standard library covering areas such as sets, relations, algebraic structures, and real analysis.

Of particular relevance to this report is Isabelle/HOL's support for probabilistic reasoning. The system includes a mature probability theory library that formalizes measure spaces, random variables, and probability distributions within the HOL framework [5]. This allows probabilistic arguments to be expressed using explicit constructions of probability spaces and measurable functions, rather than relying on informal probabilistic intuition.

Taken together, these features make Isabelle/HOL a suitable platform for the formal verification of randomized algorithms and probabilistic lemmas. It provides the logical expressiveness required to model combinatorial structures, as well as the mathematical infrastructure needed to reason rigorously about probability and randomness.

2.3 The Isolation Lemma

The Isolation Lemma, originally introduced by Mulmuley, Vazirani, and Vazirani [2], is a probabilistic principle used to ensure that, with high probability, a unique solution

exists among a family of sets. It has found widespread applications in randomized algorithms, particularly in problems involving combinatorial search, matching, and derandomization.

Informal Statement. Let U be a finite set (called the *universe*), and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subsets of U. Each element $x \in U$ is assigned a weight $w(x)$ independently and uniformly at random from $\{1, 2, \ldots, N\}$ for some integer $N \geq 1$. The total weight of a set $S \in \mathscr{F}$ is defined as

$$w(S) = \sum_{x \in S} w(x)$$

The Isolation Lemma states that, with high probability, there is a unique set in $\mathscr{F}$ that minimizes this weight function.

Theorem (Isolation Lemma) [2]: *Let U be a finite set with $|U| = n$, and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subsets. Suppose that each $x \in U$ is assigned an integer weight $w(x)$ drawn independently and uniformly at random from $\{1, 2, \ldots, N\}$. Then the probability that there exists a unique set $A \in \mathscr{F}$ that minimizes $w(S)$ is at least $1 - \frac{n}{N}$.*

$$\Pr[\exists! A \in \mathscr{F} \text{ minimizing } w(A)] \geq 1 - \frac{n}{N}$$

Proof Idea. The proof analyzes the probability that *two or more* sets in $\mathscr{F}$ attain the same minimum weight. The key observation is that such a collision can be attributed to a specific element whose randomly assigned weight causes two competing sets to tie. By bounding the probability of this event for each element and applying a union bound over the universe, one obtains the upper bound $\frac{n}{N}$ on the failure probability.

Representative Proof (after Spencer [13]). We include this classical argument as a representative proof of the Isolation Lemma for background and intuition; its fully formal treatment is developed later in the report.

Let $U = \{e_1, e_2, \ldots, e_n\}$ be a finite set of n elements, and let $\mathscr{F} \subseteq \mathscr{P}(U)$ be a non-empty family of subsets of U. Suppose each element $x \in U$ is assigned a random weight $w(x)$ uniformly from $\{1, 2, \ldots, N\}$.

For any $x \in U$, define:

$$\alpha(x) = \min_{\substack{S \in \mathscr{F} \\ x \notin S}} w(S) - \min_{\substack{S \in \mathscr{F} \\ x \in S}} w(S \setminus \{x\})$$

The quantity $\alpha(x)$ is determined by the weights of elements other than x itself, and can therefore be viewed as fixed when conditioning on all weights except $w(x)$. Since $w(x)$ is chosen independently and uniformly from $\{1, 2, \ldots, N\}$, the probability that $w(x)$ coincides with this value is at most $1/N$.

Now suppose there are two distinct sets $A, B \in \mathscr{F}$ such that both minimize the total weight $w(S) = \sum_{x \in S} w(x)$. Let $x \in A \triangle B$ (i.e., an element in the symmetric difference). Without loss of generality, suppose $x \in A \setminus B$. Then we have:

$$\alpha(x) = \min_{\substack{S \in \mathscr{F} \\ x \notin S}} w(S) - \min_{\substack{S \in \mathscr{F} \\ x \in S}} w(S \setminus \{x\})$$

A direct calculation shows that $\alpha(x) = w(x)$ for such an element x.

Thus, the event that two distinct minimizers exist implies that $w(x) = \alpha(x)$ for some $x \in U$. Since this occurs with probability at most $1/N$ for each x, a union bound over n elements yields a total failure probability of at most n/N.

Therefore, with probability at least $1 - \frac{n}{N}$, the minimum-weight set in $\mathscr{F}$ is unique.

Remarks.

- The probability bound $1 - \frac{n}{N}$ improves as N increases.
- The lemma is non-constructive: it guarantees uniqueness with high probability but does not identify the minimizing set.
- The Isolation Lemma is a central tool in randomized algorithms, e.g., in the parallel algorithm for the **Perfect Matching** problem.

This argument is presented at a high level and relies on several assumptions that are left implicit in classical presentations, including the existence of the minima involved, the independence structure of the weight assignments, and the measurability of the relevant events. Making these assumptions explicit is a central goal of the formal development in later chapters.

2.4 Parallel Algorithm for Perfect Matching

2.4.1 The Perfect Matching Problem

Let $G = (V, E)$ be an undirected graph. A *matching* in G is a set of edges such that no two edges share a common endpoint. A matching is called a *perfect matching* if every vertex in V is incident to exactly one edge in the matching.

The *Perfect Matching problem* asks whether a given graph admits a perfect matching, and in the constructive setting, how to identify such a matching. This problem occupies a central position in graph algorithms and combinatorial optimization, and has been studied extensively from both complexity-theoretic and algorithmic perspectives [3].

From a combinatorial viewpoint, the difficulty of the problem lies in the global nature of the constraint: a perfect matching must cover all vertices simultaneously, and local choices of edges may interact in complex ways. As a consequence, large graphs may admit many distinct perfect matchings, or none at all, and the presence of multiple solutions introduces significant structural symmetry.

In this report, we focus on algorithmic approaches that reason about the existence or identification of a perfect matching through indirect means, rather than by explicit enumeration of all candidate matchings. In particular, we are interested in settings where the goal is to distinguish a single solution among many combinatorially equivalent possibilities, a theme that will play a crucial role in the randomized and parallel algorithms discussed in subsequent sections.

2.4.2 Challenges in Parallel and Algebraic Approaches

One of the main difficulties in the Perfect Matching problem arises from the high degree of symmetry inherent in its solution space. A graph may admit many distinct perfect matchings, none of which is distinguished *a priori*. From an algorithmic perspective, this symmetry complicates both the decision and construction tasks, as no local criterion suffices to single out a particular solution.

This issue is particularly pronounced in the context of parallel computation. Parallel algorithms aim to exploit independence and concurrency, but symmetry among solutions often prevents independent local decisions from composing into a globally consistent matching. In the absence of a canonical choice, parallel processes may conflict or fail to converge on a single valid solution.

From a complexity-theoretic viewpoint, this symmetry barrier is a central obstacle [2], [14] in the design of efficient parallel algorithms for matching. While the Perfect Matching problem is solvable in polynomial time by classical sequential algorithms, obtaining highly parallel algorithms requires additional structure to break this symmetry in a controlled way.

Randomization provides a powerful mechanism for addressing this challenge. By introducing randomness into the problem instance, one can perturb the solution space so that, with high probability, a unique solution emerges. This transformation allows parallel procedures to reason about a single distinguished object rather than an entire symmetric family of solutions.

The Isolation Lemma formalizes this idea. By assigning random weights to the elements of the underlying structure, it ensures that, with high probability, a unique minimum-weight solution exists. In the context of the Perfect Matching problem, this means that among all perfect matchings, one becomes uniquely minimal under the random weighting. This isolation property serves as a crucial bridge between the combinatorial structure of matchings and the design of efficient randomized parallel algorithms.

In the following chapters, we make this connection precise. Chapter 3 introduces the

formal modeling of random weight assignments and their associated probability spaces, while Chapter 4 develops a fully formal proof of the Isolation Lemma. These results are then applied in Chapter 5 to support the verification of a randomized approach to the Perfect Matching problem.

2.4.3 Random Weighting and the Isolation Principle

A common strategy for breaking symmetry in combinatorial optimization problems is to introduce randomness through *random weighting*. Instead of reasoning about all feasible solutions equally, one assigns random weights to the basic elements of the structure and compare solutions by their total weight.

In the context of the Perfect Matching problem, this means assigning random weights to edges and considering the total weight of a matching as the sum of the weights of its constituent edges. While many perfect matchings may exist in the original unweighted graph, the random perturbation alters the comparison criterion among them.

The central idea behind this approach is that, under a random weighting, ties between distinct solutions become unlikely. Intuitively, two different matchings differ on at least one edge, and the random weight assigned to that edge can be expected to break the tie between their total weights. As a result, with high probability, one matching becomes strictly lighter than all others.

This intuition is formalized by the *isolation principle*. Rather than guaranteeing the existence of a solution, isolation guarantees that, among all feasible solutions, a unique minimum-weight solution emerges with high probability after random perturbation. Crucially, this uniqueness is not imposed deterministically, but is obtained as a probabilistic property of the random weight assignment.

From an algorithmic perspective, isolation transforms the structure of the problem. Instead of reasoning about a symmetric family of solutions, an algorithm can focus on a single distinguished object: the unique minimum-weight solution. This shift is particularly important in parallel computation, where global coordination among many equivalent solutions is difficult to achieve.

The Isolation Lemma provides a precise probabilistic statement of this principle. It quantifies the probability with which random weighting isolates a unique minimum-weight solution, and identifies the parameters that control the failure probability. In subsequent chapters, we formalize this lemma and show how it can be used to support the verification of randomized parallel algorithms for the Perfect Matching problem.

2.4.4 Role of the Isolation Lemma in Perfect Matching Algorithms

In randomized and parallel algorithms for the Perfect Matching problem, the Isolation Lemma plays a structural rather than computational role. It is not used to construct a matching directly, but to justify the assumption that a single, well-defined solution can be singled out from a potentially large and symmetric solution space.

Many algorithmic techniques for Perfect Matching rely on testing global properties of the solution set, such as the existence of a minimum-weight matching satisfying certain conditions. When multiple minimum solutions exist, these tests may become ambiguous or fail to compose correctly in a parallel setting. The presence of several equally optimal matchings prevents local decisions from coherently identifying a global solution.

The Isolation Lemma resolves this difficulty by guaranteeing, with high probability, the uniqueness of the minimum-weight perfect matching after random perturbation. The uniqueness allows the algorithm to treat the minimum matching as a distinguished object, eliminating ambiguity in subsequent reasoning steps. As a result, algorithmic predicates that would otherwise range over many solutions can be evaluated with respect to a single canonical one.

Importantly, algorithms do not depend on the specific mechanism by which isolation is achieved. Instead, they assume isolation as a probabilistic precondition and derive correctness from the properties that follow from uniqueness. The Isolation Lemma provides the formal justification for this assumption, linking randomized preprocessing to the correctness of higher-level algorithmic reasoning.

3 Formal Modeling of the Isolation Lemma

This chapter marks the first step in the formalization process by translating the core components of the Isolation Lemma into rigorous definitions with the Isabelle/HOL theorem proving environment. Building on the informal statement of the lemma provided in Chapter ‘’2, our objective here is to move from intuitive, pen-and-paper reasoning to a precise and structured formal model. This transition corresponds to the intermediate stage in the formalization pipeline: from Pen-and-Paper Proofs (PP) to a Formal Proof (FP) representation that is machine-readable and logically well-formed, but not yet fully proved.

We proceed by constructing the essential components of the lemma layer by layer. First, we define the underlying universe of the elements and the weight functions, and explain the rationale of choosing natural numbers for weight representation. Second, we formalize the concept of the minimizer and the condition of its uniqueness within a family of sets. Finally, we construct the product probability space using HOL-Probability Library, ensuring that the weights are distributed uniformly and independently.

These definitions act as the foundation upon which the correctness proof in Chapter 4 is built. Special attention is given to the design decisions that facilitate proof automation. In particular, we carefully handle finiteness assumptions and the construction of measure spaces, as these aspects are crucial for enabling automated reasoning in Isabelle/HOL.

Standing assumptions. Throughout this chapter, we fix a universe U and a non-empty family of sets $F \subseteq \mathscr{P}(U)$, and assume that U is finite. For the randomized construction, we fix an integer $N \geq 1$ and sample weights for all $x \in U$ independently and uniformly from $\{1, \ldots, N\}$.

3.1 Modeling Weights and Universe

Informal descriptions of randomized constructions often begin with intuitive assumptions about sets and weights, relying on natural notation without explicitly encoding their constraints. In the context of formal verification, however, these assumptions must be carefully reconstructed to support mechanized reasoning.

In the standard mathematical formulation of the Isolation Lemma, we consider a universe of elements U. A weight function, w, assigns a numerical value to each element in U. Typically, these weights are integers chosen uniformly and randomly from a range $\{1,\ldots,N\}$. The weight of a subset $S \subseteq U$ is then defined as the sum of the weights of its constituent elements:

$$w(S) = \sum_{x \in S} w(x) \tag{3.1}$$

To formalize this in Isabelle/HOL, we avoid defining an explicit "Universe" type. Instead, we represent the universe implicitly via type variables $'a$. More precisely, the universe is modeled as an explicit set $U ::' a$ `set`. The type variable $'a$ serves only as a carrier type, while all semantic statements about weights are restricted to elements $x \in U$. Although weight functions are total by definition in Isabelle, values outside the universe do not play any role in the proof. We introduce a type synonym to model the weight assignment. This approach enhances the readability of the subsequent definitions, and ensures a type safety within the proof, which is necessary for the proof assistant.

```
type_synonym 'a weight = "'a ⇒ nat"
  (* weight maps an element to a natural number *)

definition wt :: "'a weight ⇒ 'a set ⇒ nat"  where
  "wt w S = (SUM x ∈ S. w x)"
  (* sum of weight *)
```

Listing 3.1: Definition of Weight Type and Weight Function

As shown in Listing 3.1, we define the type `'a weight` as a function mapping an element of an arbitrary type `'a` to a natural number (`nat`). Based on this type, the total weight of a set S is defined using the `wt` function.

The choice of a total function is a technical requirement of Isabelle/HOL, where all functions must be defined on the entire carrier type. Semantically, all arguments and constructions in the sequel will only depend on the values of w restricted to the universe U. This separation between the carrier type and the explicit universe avoids treading the weight function as a partial function, which is not natively supported in Isabelle/HOL.

This modeling step illustrates a typical transition from pen-and-paper reasoning to formal verification. While informal proofs treat the definition of weights and subset sums as self-evident, formalization forces us to separate type-level carriers from explicitly quantified finite sets, and to express totality and domain restrictions precisely. These distinctions are not only technical, but crucial in enabling mechanized proof construction within a system like Isabelle/HOL.

DESIGN DECISION: NATURAL NUMBERS VS. INTEGERS

A critical modeling decision was the choice of the codomain for the weight function. While the original lemma is often stated using integers, we selected Isabelle's natural number type (`nat`). This decision is motivated by two factors:

- **Non-negativity:** In the context of the Isolation Lemma, weight are strictly positive (from $\{1,\ldots,N\}$). Using **nat** naturally enforces this constraint without requiring additional invariants to check for negative values.
- **Inductive Reasoning:** The `nat` type in Isabelle is powerful in induction heuristics which facilitate the automation of arithmetic proofs.

HANDLING FINITENESS

The definition of `wt` utilizes Isabelle's `sum` operator(represented as $\sum$ in the math context). It is crucial to note that in Isabelle/HOL, the summation operator is only

mathematically meaningful when the set S is finite. If S were infinite, the summation would return 0 by definition in the standard library. This finiteness condition is usually implicit in informal proofs, but must be enforce explicitly in Isabelle/HOL to avoid unintended semantics.

Consequently, throughout our formalization, we do not rely on an implicit assumption that the universe is finite, which is often overlooked in pen-and-paper proofs. As it is a mandatory step in formal verification, we explicitly enforce the assumption `finite S` or `finite U` locally in all lemmas involving weight calculations. Accordingly, all lemmas involving weight computations in this development explicitly assumes `finite U` or `finite S`, making finiteness a first-class requirement rather than an implicit convention.

3.2 Defining Minimizers and Uniqueness

A key conceptual step in understanding the Isolation Lemma is indentifying what it means for a set to be a ”minimizer”. Pen-and-paper arguments often describes this idea informally – noting that some set achieves the lowest weight – but leave the structural details implicit. To move toward a formal proof, we must refine this informal notion into a logically precise predicate.

Formally, for a family of subsets $\mathscr{F}$ and a weight function w, a set A is a minimizer if:

$$A \in \mathscr{F} \quad \text{and} \quad \forall B \in \mathscr{F},\ w(A) \leq w(B) \tag{3.2}$$

To encode this concept formally, we define a predicate `minimizer F w A` that captures both membership in $\mathscr{F}$ and optimality under the weight function. This predicate holds true if and only if the set A belongs to the family F and its weight is less than or equal to the weight of any other set B in F

```
definition minimizer :: "'a set set ⇒ 'a weight ⇒'a set
   ⇒ bool" where
  "minimizer F w A =
    (A ∈ F ∧ (∀ B ∈ F. wt w A ≤ wt w B))"
```

```
  (* A is one minimizer of F under weight w *)

definition has_unique_minimizer :: "'a set set ⇒ 'a weight
    ⇒ bool" where
  "has_unique_minimizer F w =
    (∃! A ∈ F. minimizer F w A)"
  (* then A is the unique minimizer of F under weight w *)
```

Listing 3.2: Formal Definition of a Minimizer and Uniqueness

Listing 3.2 shows how the intuitive minimizer definition is directly translated into a formal predicate suitable for mechanized reasoning. It is important to note that this definition is purely predicate-based. The predicate `minimizer F w A` does not, by itself, assert the existence or uniqueness of such a set. These properties must be established separately under additional assumptions on the family F and the weight function w. In the setting of formal proof, such distinctions are necessary to avoid implicitly assuming the existence or uniqueness of minimizers.

The Isolation Lemma goes beyond existence – its core claim concerns the **uniqueness** of the minimizer under random weights. To formalize this probabilistic uniqueness claim, we define a predicate `has_unique_minimizer`. Here, we utilize Isabelle's unique existential quantifier $\exists!$. The expression $\exists! A.P(A)$ asserts that there exists *exactly one* element A such that the predicate $P(A)$ is true. This bridges the informal idea of uniqueness with a machine-verifiable logical form.

Thus, `has_unique_minimizer F w` states that there is exactly one set A in the family F that satisfies the minimizer condition. This concise definition serves as the target statement for our main probability bound proof in the subsequent chapters.

A Useful Reformulation of Non-Uniqueness

In pen-and-paper proofs, non-uniqueness is often described informally, but in the formal setting, it is more convenient to reason with explicit witnesses. We therefore reformulate non-uniqueness using a standard logical equivalence: there is no unique

minimizer if and only if two distinct sets $A \neq B$ in $\mathscr{F}$ both satisfy the `minimizer` predicate. This reformulation enables case-based reasoning in the formal proof, where fixed witnesses simplify the structural analysis of minimizers.

3.3 The Alpha Construction

The crux of the Isolation Lemma relies on a specific value associated with each element x, commonly denoted as $\alpha(x)$. Originally introduced by Mulmuley et al. this quantity provides an intuitive threshold that governs whether x contributes to a minimum-weight set. From an informal perspective, $\alpha(x)$ reflect how sensitive the total weight is to the choice of $w(x)$. Our goal in this section is to move from this intuition toward a formal expression that supports explicit reasoning in the proof assistant.

Mathematically, $\alpha(x)$ is defined as the difference between two minimum values:

$$\alpha(x) = \min_{S \in \mathscr{F} : x \notin S} w(S) - \min_{S \in \mathscr{F} : x \in S} w(S \setminus \{x\}) \tag{3.3}$$

Intuitively, this measures the "gap" or threshold for $w(x)$. The first term represents the minimum weight achievable without using x. The second term represents the minimum "residual" weight required to complete a set that must contain x. Taken together, these two terms describe the boundary condition under which x determines the overall minimizer.

Formal Definition

In Isabelle/HOL, we formalize this using the `LEAST` operator. The two minimum values are computed independently, and their difference is taken to obtain $\alpha(x)$.

```
definition alpha :: "('a ⇒ nat) ⇒ 'a set set ⇒ 'a ⇒ nat
   " where
  "alpha w F x =
     (LEAST n. ∃ S ∈ F. x ∉ S ∧ wt w S = n)
   - (LEAST n. ∃ S ∈ F. x ∈ S ∧ wt w (S - {x}) = n)"
```

Listing 3.3: Definition of Alpha

WELL-DEFINEDNESS AND INTENDED USE

In the transition from intuitive definitions to formal ones, a key difficulty is ensuring that expressions like $\alpha(x)$ are well-defined. The use of the `LEAST` operator in Isabelle/HOL requires explicit witnesses for the corresponding minimum conditions. This is in contrast with pen-and-paper reasoning, where such existence is often assumed without justification. In our development, such existence properties are not assumed implicitly, but are discharged later as explicit proof obligations, under suitable conditions on the family $\mathscr{F}$, in particular the existence of minimizers.

At the modeling stage, $\alpha(x)$ is treated as an abstract quantity defined in terms of minimal weights. No assumptions are made at this point about which sets realize these minimum values.

In the subsequent proof, we will show that once minimizers are known to exist, the `LEAST` expressions in the definition of $\alpha(x)$ can be simplified to concrete weight values. This separation reflects a standard discipline in formal verification, where definitions are kept independent of their required assumptions.

3.4 PROBABILITY SPACE CONSTRUCTION

We now formalize the last component required to move from intuitive probabilistic assumptions (pp) to a machine-verifiable framework (fp): the construction of the probability space. Mathematically, the Isolation Lemma assumes that the weight $w(x)$ for each element $x \in U$ is chosen independently and uniformly at random from the set $\{1,\ldots,N\}$. We assume $N \geq 1$, ensuring that the set $\{1,\ldots,N\}$ is non-empty and the uniform distribution is well-defined.

Modeling: Implicit and Explicit Probability

This stage illustrates a particularly significant shift from informal reasoning to explicit construction. In standard pen-and-paper proofs, probability is often treated implicitly. We simply state "Let w be a random weight function" and proceed to calculate probabilities of events. However, in a formal proof assistant like Isabelle/HOL, there is no ambiguity allowed. We cannot merely assume randomness; we must explicitly construct the underlying probability distribution (as a PMF object).

This introduces a gap between intuition and formalization. We must define a concrete probability mass function (PMF) over the function space $U \to \{1 \ldots N\}$. To bridge this gap, we make us of the `HOL-Probability` library [5], which formalizes discrete distributions using measure theory. This library makes the probability assumptions of the Isolation Lemma explicit objects of reasoning, rather than implicit conditions.

Formalizing the Product Space

To transform the intuitive statement "weights are chosen independently and uniformly" into a formal object, we construct the joint as a product of marginal distributions:

1. **Uniform Component:** A uniform distribution over the finite set $\{1, \ldots, N\}$.
2. **Product Construction:** A product PMF over the finite domain U.

In pen-and-paper reasoning, such product distributions are usually left implicit. In our formal setting, both the uniform marginals and their combination must be constructed explicitly.

The uniform marginal distribution on $\{1, \ldots, N\}$ is implemented using `pmf_of_set` [5], which constructs a uniform PMF over any finite non-empty set.

These marginals are then combined using the `Pi_pmf` operator, which creates a product distribution over the domain U, with independent sampling at each point.

```
definition w_pmf :: "'a set ⇒ nat ⇒ 'a weight pmf" where
  "w_pmf N U =
    Pi_pmf U (0::nat) (\<lambda>_. pmf_of_set {1..N})"
```

```
(* The probability mass function for random weights *)
```

Listing 3.4: Construction of the Weight Probability Space

As shown in Listing 3.4, `w_pmf N U` defines the probability measure over total functions of type `'a => nat`, where the random choices are made independently on U. The construction uses the `Pi_pmf` operator (mathematically Π) to combine individual uniform distributions into a single product measure. This explicitly captures the **independence** assumption, ensuring that the choice of weight for any $x \in U$ does not influence others.

A technical nuance in this definition is the argument `(0::nat)` passed to `Pi_pmf`. Since functions in Isabelle/HOL are total, they must return a value even for inputs outside their logical domain ($x \notin U$). Here, `0` serves as the default value for such out-of-domain queries. While this implementation detail is necessary to satisfy Isabelle's type system, it does not affect the mathematical validity of the lemma, as our reasoning is strictly confined to the elements within U.

By modeling the randomness as a PMF object, probabilistic events become first-class objects in the logic, allowing standard measure theoretic reasoning, such as union bounds, to be carried out within Isabelle/HOL.

From Modeling to Proof. With the modeling phase complete, we now have a precise formal infrastructure: weight assignments over finite universes, a definition of minimizers and uniqueness, and an explicitly constructed product probability space.

This framework resolves the ambiguities present in informal (pen-and-paper) reasoning, where independence, randomness, and uniqueness are often stated but not justified.

In the next chapter, we transit from formal modeling to fully formal verification. Each proof step will rely on the definitions developed here and will make all logical dependencies explicit, thus completing the transition from intuitive understanding to machine-checked formal proof.

4 PROOF CONSTRUCTION

In this chapter, we construct a machine-checked proof of the Isolation Lemma in Isabelle/HOL and present the development in three layers (PP/SP/FP) to bridge informal intuition and formal verification. Unlike pen-and-paper proofs, formal verification requires every mathematical step to be made fully explicit. This includes not only the main reasoning steps, but also background obligations such as existential witnesses, measurability, independence, type correctness, and totality.

Our goal is to construct a fully formal proof (FP) of the Isolation Lemma in Isabelle/HOL. However, it is neither practical nor desirable to translate a pen-and-paper (PP) proof directly into formal code. Instead, we introduce an intermediate stage, which we refer to as the structured proof (SP) level. This stage presents a structured decomposition of the argument, designed to align with the proof obligations of Isabelle/HOL (e.g. existence, measurability, and type constraints) while remaining human-readable.

Thus, each proof component in this chapter is presented in three layers:

- **PP:** The intuitive pen-and-paper sketches from the literature.
- **SP:** A structured and detailed decomposition refined to match the proof obligations of Isabelle/HOL.
- **FP:** The corresponding formalization code in Isabelle/HOL.

We organize the whole proof as a sequence of structural reductions: We begin by establishing the existence of minimizers and by reformulating its non-uniqueness in a constructive form. Then we analyze the auxiliary quantity $\alpha(x)$, which plays a central role in isolating the dependence on individual weights. This leads to a reduction of the uniqueness failure to an explicit collision event of the form $w(x) = \alpha(x)$. Finally, we bound the probability of such events using the product probability space constructed in Chapter 3.

4.1 Existence of Minimizers

Before analyzing the uniqueness of minimizers, we must first ensure that the notion of a minimizer is well-defined. In particular, for a given family of sets $\mathscr{F}$ and a weight function w, it is necessary to establish that at least one minimizer exists whenever $\mathscr{F}$ is non-empty.

PP. In informal proofs, this step is usually omitted. Since weights are natural number and the family $\mathscr{F}$ is assumed to be non-empty, the existence of a minimizer is considered self-evident.

SP. In preparation for formalization, however, we must isolate this assumption and state it precisely, since later constructs (like $\alpha(x)$ or uniqueness predicates) depend on the existence of at least one such set.

Lemma (Existence of a Minimizer). *If $\mathscr{F} \neq \emptyset$, then there exists a set $A \in \mathscr{F}$ such that A is a minimizer of $\mathscr{F}$ under the weight function w.*

FP. This lemma is formalized as:

```
lemma exist_minimizer:
  assumes "F ≠ {}"
  obtains A where "A ∈ F" "minimizer F w A"
```

Listing 4.1: Existence of Minimizer

The proof proceeds by constructing the minimal weight value using Isabelle's `LEAST` operator. First, we extract a witness $S_0 \in \mathscr{F}$ to ensure the minimum is well-defined. We then define the minimal weight m as the least n such that some $S \in \mathscr{F}$ satisfies $wt\ w\ S = n$. A corresponding set A is then chosen to realize this minimum. Finally, we show that this A has weight at most that any other set in $\mathscr{F}$, thus satisfying the `minimizer` predicate introduced in Chapter 3.

This lemma plays a foundational role: it ensures that future uses of the `LEAST` operator or comparisons among minimizers are grounded in an actual, verifiable witness. By

discharging this assumption with a lemma, subsequent lemmas can avoid repeatedly re-establishing this basic existence condition.

4.2 REFORMULATING THE NON-UNIQUENESS

The Isolation Lemma asserts that with high probability, the weight function w yields a unique minimizer in $\mathscr{F}$. To prove this in a formal setting, we must first address the failure of uniqueness – i.e., the possibility that multiple sets in $\mathscr{F}$ achieve the minimum weight.

PP. In pen-and-paper proofs, non-uniqueness is usually described in a purely intuitive way: saying that a minimizer is ”not unique” simply means that there exist at least two different sets in $\mathscr{F}$ that both attain the minimum weight. This idea is often expressed informally by statements such as ”there are two distinct minimizers”, without any further logical unpacking.

SP. To make this intuition compatible with formal reasoning,we refine it into a positive existential formulation. Instead of negating a uniqueness statement, we characterize failure of uniqueness by the explicit existence of two distinct witnesses that both satisfy the minimizer predicate.

Concretely, the minimizer of $\mathscr{F}$ under w is not unique if and only if there exist two distinct sets $A \neq B$ in $\mathscr{F}$ such that both are minimizers. In logical terms, this corresponds to replacing a negated unique-existence statement by an equivalent existential condition of the form

$$\exists A\, B.\, A \in \mathscr{F} \;\wedge\; B \in \mathscr{F} \;\wedge\; \text{minimizer}(\mathscr{F}, w, A) \;\wedge\; \text{minimizer}(\mathscr{F}, w, B) \;\wedge\; A \neq B.$$

This reformulation eliminates the outer negation and transforms the non-uniqueness into concrete objects that can be fixed in subsequent arguments.

Moreover, if A and B are both minimizers, then they must attain the same total weight. This follows directly from the definition of the minimizer predicate: each has weight at

most that of any other set in $\mathscr{F}$, and hence in particular at most that of the other. This elementary but explicit observation will be used later when comparing the contribution of individual elements in two different minimizers.

Lemma (Non-Uniqueness via Two Minimizers). *Assume $\mathscr{F} \neq \emptyset$. Then the minimizer of $\mathscr{F}$ under w is not unique if and only if there exist two distinct sets $A, B \in \mathscr{F}$ such that both A and B are minimizers; i.e., there exist $A \neq B$ in $\mathscr{F}$ with*

$$\forall S \in \mathscr{F},\ w(A) \leq w(S) \ \wedge\ w(B) \leq w(S),$$

and in particular $w(A) = w(B)$.

FP. These facts are formalized as:

```
lemma not_unique_iff_exist_two:
  assumes "F ≠ {}"
  shows "¬ has_unique_minimizer F w ⟷
        (∃ A B. A ∈ F ∧ B ∈ F ∧
                minimizer F w A ∧
                minimizer F w B ∧
                A ≠ B)"
```

Listing 4.2: Non-Uniqueness Characterization

```
lemma minimizers_same_weight:
  assumes "minimizer F w A" "minimizer F w B"
  shows "wt w A = wt w B"
```

Listing 4.3: Minimizers Have Equal Weight

This reformulation replaces a negated uniqueness predicate by a positive existential statement that provides concrete witnesses. Once such witnesses A and B are fixed, later arguments can reason about their structural relationship, for example by analyzing elements in the symmetric difference $A \triangle B$. This will be a key step in the collision characterization developed in subsequent sections.

As an immediate consequence of this characterization, if A and B are both minimizers, they must attain the same total weight. This follows directly from the definition of the minimizer predicate and will be used implicitly in later comparisons between different minimizers.

4.3 THE ALPHA CONSTRUCTION

A central ingredient in the proof of the Isolation Lemma is the auxiliary quantity $\alpha(x)$ associated with each element $x \in U$. Intuitively, $\alpha(x)$ isolates the dependence of the total weight of a set on the individual weight $w(x)$, allowing us to reason about the effect of modifying a single coordinate.

Formally, $\alpha(x)$ is defined as the difference between two minimal weight values determined by the family $\mathscr{F}$. The first term represents the minimum total weight among all sets in $\mathscr{F}$ that do not contain x. The second term represents the minimum residual weight required to form a set in $\mathscr{F}$ that contains x, excluding the contribution of x itself. This definition was introduced by Mulmuley et al. and plays a crucial role in isolating single-coordinate events.

In our formalization, $\alpha(x)$ is defined using Isabelle's `LEAST` operator, which selects the least natural number satisfying a given predicate. We first define $\alpha(x)$ via `LEAST` to avoid committing to specific witnesses; the witness-based simplifications are derived later from the existence-of-minimizer lemmas.

In the present chapter, we analyze its properties and show how the abstract `LEAST`-based definition can be simplified in subsequent arguments, once the existence of minimizers has been derived.

4.3.1 SIMPLIFYING THE DEFINITION OF $\alpha(x)$

PP. In pen-and-paper proofs, the two minimization terms in the definition of $\alpha(x)$ are usually treated as concrete minimum values realized by suitable sets in $\mathscr{F}$. Once a minimizer exists, the least total weight among sets not containing x is simply the weight of a minimizer that avoids x. Likewise, the least residual weight among sets

containing x is obtained by removing x from a minimizer that contains it. This allows the abstract LEAST-based definition of $\alpha(x)$ to be replaced by an ordinary arithmetic expression involving concrete minimizers.

SP. To make this intuition precise, we analyze the two LEAST expressions in the definition of $\alpha(x)$ separately.

First, suppose that B is a minimizer of $\mathscr{F}$ under the weight function w, and that $x \notin B$. Since $B \in \mathscr{F}$ and attains the minimum total weight among all sets in $\mathscr{F}$, it satisfies the defining predicate of the first LEAST expression. Moreover, any other set $S \in \mathscr{F}$ with $x \notin S$ must have weight at least $w(B)$. Hence, the minimum total weight among all sets in $\mathscr{F}$ that do not contain x is attained by B, and the first LEAST expression can be shown equal to the concrete value $W(B)$.

Second, suppose that A is a minimizer of $\mathscr{F}$ and that $x \in A$. Removing x from A yields a residual set whose total weight equals $w(A) - w(x)$. Since $A \in \mathscr{F}$ and attains the minimum total weight among all sets in $\mathscr{F}$, any other set $S \in \mathscr{F}$ containing x must have residual weight at least $w(A \setminus \{x\})$. Therefore, the least value selected by the second LEAST operator coincides with the concrete value $w(A \setminus \{x\})$.

Together, these two observations eliminate both instances of the LEAST operator from the definition of $\alpha(x)$ whenever suitable minimizers are available. They reduce the abstract definition to a concrete arithmetic form that can be used directly in subsequent arguments.

FP. These simplifications are formalized by the following auxiliary lemmas in Isabelle/HOL:

```
lemma minimizer_B_eq_least_not_in:
  assumes "minimizer F w B" "x ∉ B"
  shows "(LEAST n. ∃ S. S ∈ F ∧ x ∉ S ∧ wt w S = n)
         = wt w B"
```

Listing 4.4: First LEAST Eliminated by a Minimizer

```
lemma minimizer_A_eq_least_in_residual:
```

```
  assumes "finite U" "F ⊆ Pow U" "minimizer F w A" "x ∈ A
    "
  shows "(LEAST n. ∃ S. S ∈ F ∧ x ∈ S ∧ wt w (S - {x}) = n)
        = wt w (A - {x})"
```

Listing 4.5: Second LEAST Eliminated by a Minimizer

These lemmas connect the abstract `LEAST` expressions in the definition of $\alpha(x)$ with concrete weights of minimizers. In particular, they justify replacing both minimization terms by explicit arithmetic values whenever a minimizer avoiding x or containing x is available.

As a result, the auxiliary quantity $\alpha(x)$ can be rewritten in terms of ordinary arithmetic expressions, which will be essential for proving its structural properties and for carrying out the probabilistic analysis in later sections.

4.3.2 Independence of $\alpha(x)$ from Individual Weights

PP. Intuitively, $\alpha(x)$ should not depend on the particular value of the single coordinate $w(x)$. Changing the value of $w(x)$ shifts the total weight of every set containing x by the same offset, while leaving the relative ordering of weights determined by the remaining elements unchanged. Hence the two minima defining $\alpha(x)$ – one over sets excluding x, and one over residual weights of sets including x – remain invariant under modifications of $w(x)$.

SP. We now formulate this independence property as a formal lemma, which will serve as a key technical interface for the probabilistic argument in the next section.

Lemma (Independence of $\alpha(x)$). *Let w and w' be two weight functions such that $w(y) = w'(y)$ for all $y \in U \setminus \{x\}$. Then $\alpha(x)$ computed with respect to w is equal to $\alpha(x)$ computed with respect to w'.*

To prove this lemma, we compare the two definitions of $\alpha(x)$ under w and w'. The core observation is that for any set $S \subseteq U$ with $x \notin S$, the total weight $wt\ w\ S$ depends only on values of w on $S \subseteq U \setminus \{x\}$, and therefore equals $wt\ w'$§. The same reasoning

applies to residual sets of the form $S - \{x\}$, since $S - \{x\} \subseteq U \setminus \{x\}$.

Using this congruence of weights, we show that the predicates appearing in the two LEAST expressions defining $\alpha(x)$ are equivalent under w and w'.

$$\exists S \in F.\, x \notin S \wedge wt\ w\ S = n \quad \Longleftrightarrow \quad \exists S \in F.\, x \notin S \wedge wt\ w'\ S = n,$$

and similarly for the residual term

$$\exists S \in F.\, x \in S \wedge wt\ w\ (S - \{x\}) = n \quad \Longleftrightarrow \quad \exists S \in F.\, x \in S \wedge wt\ w'\ (S - \{x\}) = n.$$

Therefore, the two LEAST values coincide under w and w'. Unfolding the definition of $\alpha(x)$ then yields the desired conclusion of the lemma, namely $\alpha_w(x) = \alpha_{w'}(x)$.

FP. This lemma is formalized as `alpha_independent_of_coord` in Isabelle/HOL:

```
lemma alpha_independent_of_coord:
  assumes "finite U" "F ⊆ Pow U" "x ∈ U"
  shows "∀ w w'. (∀ y. y ∈ U - {x} ⟶ w y = w' y)
        ⟶ alpha (w :: 'a ⇒ nat) F x = alpha w' F x"
```

Listing 4.6: Independence of alpha(x) from the x-coordinate

This lemma is proved by establishing weight congruence on sets contained in $U - \{x\}$, transferring this congruence to the defining predicates of the two LEAST expressions in $\alpha(x)$, and finally unfolding `alpha_def`.

4.3.3 Collision characterization via $\alpha(x)$

PP. We now establish the key structural link between the failure of uniqueness and the auxiliary quantity $\alpha(x)$.

Assume that the minimizer of $\mathscr{F}$ under w is not unique. Then there exist two distinct minimizers $A, B \in \mathscr{F}$. Choose an element $x \in A \setminus B$. Since A and B are minimizers, their total weights coincide:

$$w(A) = w(B).$$

Moreover, in the definition of $\alpha(x)$, the minimum over sets not containing x is realized by B, while the minimum over residual weights of sets containing x is realized by $A \setminus \{x\}$. Hence,

$$\alpha(x) = w(B) - w(A \setminus \{x\}) = w(A) - \big(w(A) - w(x)\big) = w(x).$$

Therefore, non-uniqueness implies the existence of an element $x \in U$ such that $w(x) = \alpha(x)$.

SP. The pen-and-paper argument above already matches closely the structure of the formal proof. The only additional ingredients required for formalization are the explicit side conditions ensuring that the two `LEAST` expression in the definitions of $\alpha(x)$ are well-defined and can be eliminated.

Concretely, we assume that the universe U is finite and that the family $\mathscr{F}$ is a collection of subsets of U. Under these assumptions, the two auxiliary lemmas established in Section 4.3 apply: the first shows that the `LEAST` term over sets not containing x is realized by B whenever $x \notin B$, and the second shows that the residual `LEAST` term over sets containing x is realized by $A \setminus \{x\}$ whenever $x \in A$.

Together with the lemma stating that any two minimizers have equal total weight, these results allow the formal proof to mirror the pen-and-paper derivation step by step. No additional combinatorial insight is required at this stage; the formalization consists entirely in assembling previously established lemmas and discharging the necessary finiteness and membership conditions.

FP. This reasoning is captured in Isabelle/HOL by the following lemma:

```
lemma two_minimizers_alpha_eq:
  assumes "finite U" "F ⊆ Pow U"
          "A ∈ F" "B ∈ F"
          "minimizer F w A" "minimizer F w B"
          "x ∈ A - B"
  shows   "alpha w F x = w x"
```

Listing 4.7: Collision Witnessed by a Single Coordinate

The proof unfolds the definition of $\alpha(x)$ and eliminates both `LEAST` expressions using the auxiliary lemmas from Section 4.3. The equality of weights $w(A) = w(B)$ follows from the fact that A and B are both minimizers. Finally, the standard splitting identity $w(A) = w(A \setminus \{x\}) + w(x)$ is used to derive the desired conclusion $\alpha(x) = w(x)$.

4.4 Probabilistic Analysis and Final Bound

PP. We now complete the proof of the Isolation Lemma by bounding the probability that the minimizer of $\mathscr{F}$ is not unique under a random weight assignment. Throughout this section, weights are sampled according to the product probability space constructed in Chapter 3.

By the collision characterization established in Section 4.3.3, the event that the minimizer is not unique can be reduced to a union of single-coordinate events, following the counting strategy established in the literature [13]:

$$\{\,\text{the minimizer is not unique}\,\} \subseteq \bigcup_{x \in U} \{\, w(x) = \alpha(x) \,\}.$$

Thus, it suffices to bound the probability of each event $\{\, w(x) = \alpha(x) \,\}$.

Fix an arbitrary element $x \in U$. By the independence property proved in Section 4.3.2, the auxiliary quantity $\alpha(x)$ depends only on the weights assigned to elements in $U \setminus \{x\}$. Consequently, once all other weights have been fixed, $\alpha(x)$ becomes a constant with respect to the remaining random choice of $w(x)$.

Under the product probability space, the random variable $w(x)$ is distributed uniformly over the finite set $\{1, \ldots, N\}$ and is independent of all other coordinates. Therefore, the probability that $w(x)$ attains any particular value is exactly $1/N$, and in particular

$$\Pr\big[\, w(x) = \alpha(x) \,\big] \leq \frac{1}{N}.$$

Since the universe U is finite, we may apply the union bound:

$$\Pr\big[\text{the minimizer is not unique}\big] \leq \sum_{x \in U} \Pr\big[\, w(x) = \alpha(x) \,\big] \leq \frac{|U|}{N}.$$

Equivalently, the probability that the minimizer is unique satisfies

$$\Pr\big[\text{the minimizer is unique}\big] \geq 1 - \frac{|U|}{N}.$$

This inequality constitutes the statement of the Isolation Lemma.

SP. The formal proof follows the same high-level structure as the pen-and-paper argument, but it must expose the proof obligations that are implicit informally.

To obtain an explicit probability bound inside Isabelle/HOL, the event $\{\, w(x) = \alpha(x) \,\}$ is decomposed into a finite union over $k \in \{1, \ldots, N\}$:

$$\{\, w(x) = \alpha(x) \,\} \;=\; \bigcup_{k=1}^{N} \{\, \alpha(x) = k \;\wedge\; w(x) = k \,\}.$$

This technical decomposition allows the application of independence at the level of random variables together with uniform point-probability bounds.

To obtain a multiplicative bound on each summand, the independence between $w(x)$ and $\alpha(x)$ must be established at the level of random variables. This is achieved by exploiting the product structure of the underlying PMF. The restriction of a sample function to $\{x\}$ and to $U \setminus \{x\}$ yields two independent components. Since $w(x)$ depends only on the first component, while $\alpha(x)$ depends only on the second (by the independence-of-α lemma), their compositions are independent random variables. This yields

$$\Pr[\, \alpha(x) = k \wedge w(x) = k \,] = \Pr[\, \alpha(x) = k \,] \cdot \Pr[\, w(x) = k \,].$$

Finally, the construction of $\alpha(x)$ via minimum operators over finite families requires auxiliary measurability arguments. Finite sums of coordinates are shown to be

measurable, predicates of the form $\exists S \in F. \cdots$ are measurable over the product space, and the resulting LEAST expressions preserve measurability. These facts ensure that all events appearing in the above decomposition are measurable and that the standard independence and union-bound lemmas can be applied without further side conditions.

FP. In Isabelle/HOL, the above structural steps are realized by the following core lemmas and theorems:

(a) Coordinate support and point probabilities. We first show that for any $x \in U$, the random variable $w(x)$ takes values almost surely in $\{1,\ldots,N\}$. This yields the following support lemma:

```
lemma w_pmf_component_in_range:
  assumes "x ∈ U" "N ≥ 1" "finite U"
  shows "set_pmf (map_pmf (λ w. w x) (w_pmf N U)) ⊆ {1..N
     }"
```

Listing 4.8: Coordinate Support

Based on this, we derive the uniform point probability formula for each $k \in \{1,\ldots,N\}$ (appearing as a subgoal in the main proof):

```
measure_pmf.prob (w_pmf N U) {w. w x = k} = 1 / real N
```

Listing 4.9: Point Probability of One Coordinate

(b) Measurability lemmas (measurability of α). To legitimately apply probabilistic reasoning, we must show that the auxiliary quantity $\alpha(x)$ is a measurable random variable. The formalization culminates in:

```
lemma alpha_measurable:
  assumes "finite U" "F ⊆ Pow U" "x ∈ U"
  shows "(λ f. alpha (λ y. if y ∈ U-{x} then f y else 0) F
     x)
         ∈ (Π_M (U-{x}) (λ _. count_space UNIV)) →_M (
            count_space UNIV)"
```

Listing 4.10: Measurability of Alpha

This proof relies on measurability of finite sums, measurability of finite existential predicates, and the measurability theorem for the `LEAST` operator.

(c) Independence lemma. Next, we lift coordinate-wise independence to random-variable independence by showing that $w(x)$ is independent of $\alpha(x)$, which depends only on the remaining coordinates:

```
prob_space.indep_var (measure_pmf (w_pmf N U))
  (count_space UNIV) (λ w. w x)
  (count_space UNIV) (λ w. alpha w F x)
```

Listing 4.11: Independence of Coordinate and Alpha

This step combines restriction lemmas for product measures with the previously established lemma `alpha_independent_of_coord`, ensuring that $\alpha(x)$ depends solely on the weights assigned to $U - \{x\}$.

(d) Main theorem: Isolation Lemma. Finally, combining the single-coordinate probability bound with the union bound yields the formalized version of the Isolation Lemma:

```
theorem isolation_lemma_main:
  fixes U :: "'a set" and F :: "'a set set" and N :: nat
  assumes "finite U" "F ⊆ Pow U" "F ¬eq {}" "N ≥ 1"
  shows "measure_pmf.prob (w_pmf N U){w.
     has_unique_minimizer F w}
         ≥ 1 - real(card U) / real N"
```

Listing 4.12: Main Theorem: Isolation Lemma

This theorem completes the formal derivation from the structural collision event to the global lower bound on the probability of uniqueness.

Taken together, these lemmas bridge the gap between the informal probabilistic argument and its fully formal counterpart. They make explicit the two technical ingredients that are implicit in pen-and-paper proofs: the measurability of $\alpha(x)$ and its independence from the coordinate $w(x)$. With these in place, the Isolation Lemma follows directly by applying the union bound inside Isabelle/HOL's probability framework.

4.5 Chapter Summary

In this chapter, we presented a fully formal proof of the Isolation Lemma in Isabelle/HOL, by systematically refining a classical proof into structured proof steps and finally machine-checked Isabelle/HOL code. We reformulated non-uniqueness as the existence of two distinct minimizers, introduced the auxiliary quantity $\alpha(x)$, and showed how it captures the precise collision event responsible for ties. A key technical ingredient is showing that that $\alpha(x)$ is independent of the individual weight $w(x)$, which enables a reduction to single-coordinate probabilistic events. Finally, by establishing measurability, independence, and a uniform point probability bound within a product probability space, we derived the standard $1 - |U|/N$ uniqueness bound in a fully mechanized manner.

5 Perfect Matching Problem

5.1 Introduction to Randomized Parallel Matching

The Perfect Matching problem is a fundamental problem in combinatorial optimization and complexity theory. Formally, given a bipartite graph $G = (U \cup V, E)$ where $|U| = |V| = n$. a *matching* is defined as a set of edges $M \subseteq E$ such that no two edges in M share a common vertex. A matching M is said to be *perfect* if it contains exactly n edges, effectively covering every vertex in the graph.

While efficient sequential algorithms for finding perfect matchings, such as the augmenting path method, have been well-established, developing an efficient parallel algorithm remains a significant challenge. This gap between sequential and parallel complexity is not merely a matter of implementation, but is rooted in the structural properties of the problem.

5.1.1 Decision versus Construction

In complexity theory, there is a notable distinction between the *decision* version of a problem (answering if a matching exists) and the *construction* version(outputting the edges of the matching). For the perfect matching problem, the decision version can be solved in parallel by calculating the determinant of the Tutte matrix, which belongs to the complexity class **NC**.

However, constructing the matching itself is significantly harder in a parallel setting. The standard sequential approach of greedily picking edges or finding augmenting paths is inherently serial, as each step depends on the state resulting from the previous step. This sequential dependency makes it difficult to distribute the workload across multiple processors effectively.

5.1.2 THE SYMMETRY BARRIER AND COORDINATION

The primary obstacle to parallelizing the search for a perfect matching is known as the *Symmetry Barrier* [15]. In many graphs, there can be an exponential number of distinct perfect matchings. In a sequential setting, an algorithm can easily break this symmetry by picking the first valid matching it finds.

In a parallel environment, multiple processors must work on different edges simultaneously. If the graph contains multiple perfect matchings, say M_1 and M_2, one processor might select an edge from M_1, while another selects an edge from M_2. Because these edges may not belong to the same global solution, their union is unlikely to form a valid perfect matching.

Without a mechanism to ensure that all processors converge on the *same* solution, parallel coordination becomes impossible.

5.1.3 ISOLATION AS A SOLUTION

To overcome the symmetry barrier, we utilize the **Randomized NC (RNC)** strategy proposed by Mulmuley, Vazirani, and Vazirani [15]. The core idea is to transform the symmetric solution space into an asymmetric one by introducing randomness.

By assigning random weights to each edge, we ensure that different perfect matchings likely have different total weights. The **Isolation Lemma**, formally proved in Chapter 4, provides the mathematical guarantee that with high probability, there is a *unique* minimum-weight perfect matching.

This uniqueness is the foundation of the parallel algorithm: if there is only one target solution, every processor can independently decide if its edge is part of that specific matching. The need for inter-processor communication is eliminated, and the symmetry is broken. In the following sections, we formalize how this uniqueness allows us to ”extract” the matching using algebraic methods.

5.2 Algebraic Modeling: The Tutte Matrix

To design an efficient parallel algorithm, we must transform the combinatorial search for matchings into a problem that can be solved using parallel linear algebra. The standard approach utilizes the *Tutte matrix* [2] (often referred to as the Edmonds matrix in the bipartite case) to encode the graph structure into a matrix form.

5.2.1 Matrix Construction in Isabelle/HOL

In our formalization, we consider a bipartite graph $G = (V, E)$ with n vertices on each side. We represent the weight of an edge (i, j) as $w_{i,j}$. In the `bipartite_graph` locale, the matrix D is constructed such that each entry $D_{i,j}$ encodes both the existence of an edge and its weight:

```
definition D_entry ::
    "nat ⇒ nat ⇒ int" where
    "D_entry i j = (if (i,j) : E then (2::int) ^ (w i j)
       else 0)"
```

Listing 5.1: Matrix Entry Definition

The choice of $2^{w_{i,j}}$ as the matrix entry, as proposed in [2], [15] is a critical design decision to prevent algebraic cancellation. By using powers of 2, the total weight of a perfect matching becomes the exponent in the Leibniz expansion of the determinant. This algebraic encoding allows the determinant to ”store” information about all feasible matchings simultaneously within its binary representation.

5.2.2 Determinant and Perfect Matchings

A perfect matching in a bipartite graph can be formally defined as a permutation p of the set $\{0, \ldots, n-1\}$ such that every pair $(i, p(i))$ is a valid edge in E. Our formal definition in Isabelle/HOL captures this requirement:

```
definition is_perfect_matching ::
    "(nat ⇒ nat) ⇒ bool" where
  "is_perfect_matching p ⟷
```

```
    (p permutes {0..<n} ∧ (∀ i < n. (i, p i) : E))"
```

Listing 5.2: Perfect Matching Definition

The determinant of matrix D, defined via the Leibniz formula, serves as the primary tool for isolation:

$$det(D) = \sum_{p \in S_n} \text{sgn}(p) \prod_{i=0}^{n-1} D_{i,p(i)} \tag{5.1}$$

By substituting the definition of $D_{i,j}$, any term in the sum is non-zero if and only if the permutation p corresponds to a valid perfect matching. For such a permutation, the contribution to the determinant is:

$$\text{sgn}(p) \cdot \prod_{i=0}^{n-1} 2^{w_{i,p(i)}} = \text{sgn}(p) \cdot 2^{\sum w_{i,p(i)}} = \text{sgn}(p) \cdot 2^{weight(p)} \tag{5.2}$$

If p does not represent a perfect matching, at least one $D_{i,p(i)}$ is 0, causing the entire product to vanish. This mapping ensures that the determinant is a weighted sum of all perfect matchings, providing the necessary algebraic foundation for the subsequent extraction of the unique minimum matching.

5.3 The Weight-to-Multiplicity Bridge

The core of the parallel matching algorithm lies in the relationship between the determinant of the Tutte matrix and the weights of the perfect matchings it represents. Once the Isolation Lemma ensures that the minimum-weight perfect matching p_{min} is unique, the determinant $det(D)$ effectively becomes a binary encoding of the matching weights.

5.3.1 The Non-Cancellation Principle

In the Leibniz expansion of $det(D)$, each perfect matching p contributes a term of the form $\text{sgn}(p) \cdot 2^{weight(p)}$. A potential concern in this algebraic sum is that terms with

opposite signs might cancel each other out. However, because the weights are stored as exponents of 2, the term with the smallest exponent cannot be cancelled by any combination of terms with larger exponents.

Formally, if W_0 is the unique minimum weight, $det(D)$ can be expressed as:

$$det(D) = \text{sgn}(p_{min})2^{W_0} + \sum_{p \in P \setminus \{p_{min}\}} \text{sgn}(p)2^{weight(p)} \tag{5.3}$$

Since for all other matchings p, $weight(p) > W_0$, every term in the summation is a multiple of 2^{W_0+1}. Therefore, the entire determinant is a multiple of 2^{W_0} but not of 2^{W_0+1}, ensuring that the lowest non-zero bit of the sum is preserved.

5.3.2 Formalization via Multiplicity

To extract this minimum weight in Isabelle/HOL, we utilize the `multiplicity` function from the number theory library. The multiplicity of a prime k in an integer x is the largest power of k that divides x. In our binary context, `multiplicity 2 det_D` yields exactly W_0.

We formalized this connection through several key lemmas. First, `multiplicity_2_abs` establishes the base case for powers of 2:

```
lemma multiplicity_2_abs:
  fixes x :: int
  assumes "abs x = 2 ^ k"
  shows "multiplicity 2 x = k"
```

Listing 5.3: Multiplicity of Power of 2

Then, we established the main theorem relating the determinant to the unique minimum weight:

```
lemma unique_det_power_of_weight:
  assumes "unique_min_weight_condition"
  shows "multiplicity 2 det_D = min_weight_val"
```

Listing 5.4: Multiplicity of Determinant

This result is proven in Isabelle by applying `multiplicity_eqI`, which requires showing that 2^{W_0} divides the sum and 2^{W_0+1} does not. The proof handles the summation by splitting it into the unique minimum term and the remaining "heavy" terms, formalizing the intuitive non-cancellation argument presented earlier.

5.4 Parallel Edge Extraction Logic

The final stage of the parallel algorithm is to identify which edges $(i, j) \in E$ constitute the unique minimum-weight perfect matching p_{min}. Crucially, this step must be performed in parallel, meaning each edge must be testable independently of others.

5.4.1 Minors and Subgraph Matchings

For any edge (i, j) we consider the minor $D_{i,j}$, which is the submatrix obtained by deleting the i-th row and j-th column of D. Algebraically, the determinant of this minor represents the sum of all perfect matchings in the subgraph $G \setminus \{u_i, v_j\}$. If an edge (i, j) belongs to a perfect matching p in G, then $p \setminus \{(i, j)\}$ must be a perfect matching in the reduced graph.

In our formalization, we define `minor_sum i j` to represent the contribution of all matchings containing the edge (i, j). This relationship is captured by the lemma `minor_sum_decomposition`:

```
lemma minor_sum_decomposition:
  assumes "i < n" "j < n" "(i,j) : E"
  shows "minor_sum i j = D_entry i j * cofactor_sum i j"
```

Listing 5.5: Minor Sum Decomposition

This decomposition allows us to isolate the contribution of a single edge weight from the rest of the matching structure.

5.4.2 The Parallel Decision Predicate

To determine if $(i, j) \in p_{min}$, we compare the minimum weight of matchings containing (i, j) with the global minimum weight W_0. If (i, j) is part of the unique minimum

matching, the minimum weight in the subgraph plus the edge weight $w_{i,j}$ must equal the global minimum weight.

This core division logic can be expressed by the following identity:

$$w(i,j) + \text{multiplicity}_2(\det(D_{i,j})) = min_weight_val \tag{5.4}$$

We formalized this as a necessary and sufficient condition in the theorem `edge_test_iff_correct`:

```
theorem edge_test_iff_correct:
  assumes p_min_def:
    "is_perfect_matching p_min"
    "matching_weight p_min = min_weight_val"
  assumes p_min_unique:
    "∀ q. is_perfect_matching q ∧ matching_weight q
            = min_weight_val ⟶ q = p_min"
  shows "p_min i = j ⟷
         (cofactor_sum i j ≠ 0 ∧ w i j + multiplicity 2 (
            cofactor_sum i j) = min_weight_val)"
```

Listing 5.6: Correctness of Edge Extraction

This theorem provides the formal justification for the parallel algorithm. Each edge (i,j) can be tested independently by computing the determinant of its corresponding minor. If the weight condition holds, the edge is guaranteed to be part of the unique matching p_{min}. Since determinant calculation is in **NC**, the entire matching can be constructed in parallel time.

5.5 Probabilistic Correctness via Isolation Lemma

The formal verification of the parallel matching algorithm is completed by establishing its probabilistic success guarantee. While Chapter 4 provides a general proof of the

Isolation Lemma for any finite family of sets, we must now specialize this result to the domain of perfect matchings in a bipartite graph.

5.5.1 Mapping Graphs to Sets

To apply the Isolation Lemma, we first define a formal mapping between graph-theoretic structures and the set-theoretic framework. The universe U is identified with the set of edges E, and the family $\mathscr{F}$ is defined as the collection of all edge sets that constitute a valid perfect matching:

```
definition perfect_matching_family ::
    "nat ⇒ (nat * nat) set ⇒ ((nat * nat) set) set" where
    "perfect_matching_family n E =
      {matching_to_set n p ∨ p.
       p permutes {0..<n} ∧ (∀ i < n. (i, p i) : E)}"
```

Listing 5.7: Perfect Matching Family

A critical step in this bridge is ensuring that the weight of a matching in the graph model is identical to the weight of the corresponding set in the isolation model. We established this via the lemma `weight_equivalence`. Furthermore, the lemma `matching_to_set_inj` ensures that each permutation corresponds to a unique set of edges, preventing any unintended collisions in the probability space.

5.5.2 Final Probability Bound

The equivalence between the graph-theoretic unique minimum condition and the set-theoretic `has_unique_minimizer` predicate is established in the lemma `pm_iso_equivalence`. This allows us to lift the general isolation bound to the specific case of bipartite matching.

The theorem `isolation_lemma_for_perfect_matching` provides the machine-checked lower bound on the success probability:

```
theorem isolation_lemma_for_perfect_matching:
  fixes N :: nat
```

```
  assumes "N ≥ 1"
  assumes "∃ p. is_perfect_matching p"
  assumes "E ≤ {0..<n} * {0..<n}"
  shows "measure_pmf.prob (w_pmf N E)
      {w. unique_min_weight_condition (λ u v. w (u,v)) E}
      ≥ 1 - (real (card E) / real N)"
```

Listing 5.8: Isolation Lemma for Perfect Matching

This theorem completes the formal derivation. For a graph with edge set E and weights chosen uniformly from $\{1,\ldots,N\}$, the probability that the algorithm identifies a unique matching is at least $1-|E|/N$. This result provides the formal justification for using randomized weighting to achieve parallel coordination in the Perfect Matching problem.

5.6 Formalization Challenges in Isabelle/HOL

The formalization of randomized parallel matching involves several technical challenges that are absent in informal presentations. These difficulties arise primarily from the strict requirement for type safety and the integration of disparate mathematical libraries within Isabelle/HOL.

5.6.1 Bridging Type Systems: Int versus Nat

A significant hurdle during implementation was the reconciliation of different numeric types. In Isabelle/HOL, the `multiplicity` function is defined over the `int` type within the number theory library. However, the `Determinants` library often operates over more general algebraic structures.

Since the edge weights are naturally non-negative, we initially modeled them as `nat`. Yet, the determinant of a matrix with `sign p` (which can be -1) requires the codomain to be `int`. This necessitated the use of explicit type casting and lemmas to prove

that properties like multiplicity are preserved. This technical detail required formal justification via lemmas such as `multiplicity_2_not_dvd_Suc`.

5.6.2 Library Integration and Automation

Another challenge involved the integration of the `Permutations` library with the `Probability` library. To prove the final probability bound, we had to ensure that the combinatorial structure could be measured within the product probability space.

A major effort was spent on the lemma `unique_subset_sum_power_of_weight`. In an informal setting, one might simply state that higher powers of 2 do not affect the lowest bit. In Isabelle, this requires a rigorous proof to show that a sum of multiples of 2^{k+1} remains a multiple of 2^{k+1}, thus not interfering with a single term of value 2^k. The use of structured proof steps allowed us to manage this complexity by breaking the summation logic into manageable sub-goals.

5.7 Chapter Summary

In this chapter, we have presented a complete formal verification of a randomized parallel algorithm for the Perfect Matching problem. By constructing the problem as a bipartite graph within Isabelle/HOL, we demonstrated how the abstract probabilistic guarantees of the Isolation Lemma can be translated into concrete algebraic properties of the Tutte matrix.

We formally established that by encoding edge weights as powers of 2, the determinant of the matrix becomes a structured representation of all possible matchings. The core of our verification relied on the ”Weight-to-Multiplicity” bridge, where we proved that the multiplicity of 2 in the determinant identifies the exact weight of the unique minimum-weight matching. Furthermore, we verified the parallel edge extraction logic, proving that each edge can independently determine its membership in the matching by calculating its corresponding minor. This chapter completes the transition from the foundational probability theory established in Chapter 4 to a high-level application in randomized algorithm analysis, achieving a

fully machine-checked proof of correctness for a Randomized NC (RNC) algorithm algorithm.

6 Conclusion and Future Work

6.1 Summary of Research Achievements

This report has presented a comprehensive formal verification of a Randomized Parallel Matching algorithm, centered around a machine-checked proof of the Isolation Lemma in Isabelle/HOL. By navigating the transition from pen-and-paper probabilistic intuition to structured formal logic, we have established a rigorous foundation for reasoning about randomized algorithmic constructions.

The primary contributions of this work are:

- **Rigorous Foundation for Isolation**: We constructed a complete product probability space and proved the $1 - |U|/N$ success bound, making all assumptions about measurability and independence explicit.
- **Algorithmic Correctness**: We successfully applied the Isolation Lemma to verify the "Weight-to-Multiplicity" bridge and the parallel edge extraction logic using the Tutte matrix and its minors.
- **Bridging the Formalization Gap**: We employed a layered PP/SP/FP approach to connect informal folklore-style proofs with fully formal machine-checked code.

6.2 Methodological Lessons and Reflections

The process of formalizing the Isolation Lemma and its application to the Perfect Matching problem has yielded several methodological insights into the nature of randomized algorithm verification.

6.2.1 The Ontology of Randomness: From Intuition to PMF Objects

A primary reflection from the modeling phase in Chapter 3 is that randomness must be treated as a first-class mathematical object rather than an implicit environmental property. In classical RNC analysis, probabilistic intuition often obscures subtle dependencies. By explicitly constructing product probability spaces using the `Pi_pmf` constructor, we transitioned from "vague" randomized processes to machine-verifiable measures. This rigor is essential to ensure that every assumption about independence is logically grounded.

6.2.2 Structural Bridging: The Utility of the SP Layer

As demonstrated in Chapter 4, the introduction of the **Structured Proof (SP)** layer served as a vital cognitive bridge. While SP is not a logical innovation per se, it acts as a diagnostic tool that decomposes "compressed" folklore arguments into granular steps. This methodology revealed that many "obvious" steps in the literature require non-trivial formal justification–specifically the measurability of the threshold $\alpha(x)$, a property consistently treated as implicit in pen-and-paper proofs.

6.2.3 Representation and Consistency: Permutations versus Sets

In Chapter 5, we encountered the challenge of bridging the **representation gap** between graph-theoretic permutations and set theoretic edge families. Formalizing this as an injective mapping was a critical lesson: the correctness of an algorithm depends not only on the logical steps but also on the consistency of data representations. Without a formal proof of injectivity, the probabilistic bounds could not be safely translated to the bipartite matching model.

6.2.4 The Hidden Burden of Rigor: Type Safety and Measurability

Finally, the reconciliation of type systems (specifically `nat` and `int`) and the requirement for measurability proofs provided a rigorous safety layer. While explicit

type casting and measurability obligations added to the proof's complexity, they eliminated the risk of subtle errors. This underscores that in formal verification, there is no "minor detail", every element of reasoning must be subjected to the highest level of scrutiny.

6.3 Future Work and AFP Submission

The results of this project are currently being prepared for submission to the **Archive of Formal Proofs (AFP)**. As a curated collection of Isabelle/HOL developments, the AFP entry will provide a reusable library for the Isolation Lemma, enabling future researchers to verify more complex randomized data structures and algorithms based on the results submitted and archived.

Future research could extend this work in several directions:

- **Verification of Diverse RNC Problems:** The algebraic approach used for Perfect Matching can be adapted to other problems in the **Randomized NC (RNC)** class. For instance, the isolation principle is equally vital for parallel algorithms solving the *Matrix Rank* or *All-Pairs Shortest Paths* problems.
- **Expanding the Randomized Algorithm Library**: The current formalization can serve as a foundation for verifying other probabilistic building blocks, such as **Chernoff bounds** [16] or **Hashing lemmas** [17], creating a more comprehensive toolkit for randomized algorithm analysis in Isabelle/HOL.
- **Application to Other Combinatorial Structures**: While this work focuses on bipartite graphs, the same weighting logic can be tested on simpler combinatorial structures, such as spanning trees or specific subset-sum problems, to evaluate the lemma's versatility.
- **Proof Automation for Probabilistic Reasoning**: Future efforts could focus on developing specialized Isabelle strategies to automate tedious formalization tasks, such as measurability proofs and summation index-shifting, reducing the manual effort required for randomized proofs.

In conclusion, this work demonstrates that even subtle probabilistic arguments can be subjected to the highest level of logical scrutiny. By digitalizing these foundational results, we contribute to the ongoing effort of building a verified library of randomized algorithm theory.

REFERENCES

[1] R. Motwani and P. Raghavan, *Randomized Algorithms*. Cambridge, UK: Cambridge University Press, 1995.

[2] K. Mulmuley, U. V. Vazirani, and V. V. Vazirani, "Matching is as easy as matrix inversion," *Combinatorica*, vol. 7, no. 1, pp. 105–113, 1987. DOI: 10.1007/BF02579206

[3] L. Lovász, "On determinants, matchings and random algorithms," in *Fundamentals of Computation Theory (FCT '79)*, vol. 79, Berlin, Heidelberg: Springer, 1979, pp. 565–574.

[4] T. Nipkow, L. C. Paulson, and M. Wenzel, *Isabelle/HOL: A Proof Assistant for Higher-Order Logic* (Lecture Notes in Computer Science). Springer-Verlag, 2002, vol. 2283. DOI: 10.1007/3-540-45949-9

[5] The Isabelle Development Team, "The Isabelle/HOL probability library," https://isabelle.in.tum.de/dist/library/HOL/HOL-Probability/document.pdf, Accessed: 2026-03-08, 2026.

[6] The QED Group, "The QED manifesto," in *Automated Deduction—CADE-12*, A. Bundy, Ed., Berlin, Heidelberg: Springer Berlin Heidelberg, 1994, pp. 238–251. DOI: 10.1007/3-540-58156-1_17

[7] Y. Chen, "Quantum algorithms for lattice problems," Cryptology ePrint Archive, Paper 2024/555, 2024. https://eprint.iacr.org/2024/555.

[8] K. Kosaian, Y. K. Tan, and K. Y. Rozier, "Formalizing coppersmith's method in isabelle/hol," in *Intelligent Computer Mathematics*, A. Kohlhase and L. Kovács, Eds., Cham: Springer Nature Switzerland, 2024, pp. 127–145, ISBN: 978-3-031-66997-2.

[9] E. Karayel, S. J. Watt, D. Khu, K. S. Meel, and Y. K. Tan, "Verification of the CVM Algorithm with a Functional Probabilistic Invariant," in *16th International Conference on Interactive Theorem Proving (ITP 2025)*, Y. Forster and C. Keller, Eds., ser. Leibniz International Proceedings in Informatics (LIPIcs), vol. 352, Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Infor-

matik, 2025, 34:1–34:20, ISBN: 978-3-95977-396-6. DOI: 10.4230/LIPIcs.ITP.2025.34

[10] V. V. Vazirani, *Approximation Algorithms*. Springer, Berlin, Heidelberg, 2001. DOI: 10.1007/978-3-662-04565-7

[11] X. Leroy, "Formal verification of a realistic compiler," *Communications of the ACM*, vol. 52, no. 7, pp. 107–115, 2009. DOI: 10.1145/1538788.1538814

[12] G. Gonthier, "A computer-checked proof of the Four Colour Theorem," Technical Report, Microsoft Research Cambridge, 2005.

[13] J. Spencer, "Probabilistic methods," in *Handbook of Combinatorics*, R. L. Graham, M. Grötschel, and L. Lovász, Eds., vol. 2, Elsevier Science, 1995, pp. 1786–1817.

[14] A. Wigderson, "NL/poly $\subseteq \oplus$L/poly," in *Proceedings of IEEE 9th Annual Conference on Structure in Complexity Theory*, 1994, pp. 59–62. DOI: 10.1109/SCT.1994.315817

[15] A. Gupta, "15-859m: Randomized algorithms, lecture 10: Polynomial identity testing and parallel matchings," Carnegie Mellon University, Computer Science Department, 2011. https://www.cs.cmu.edu/~avrim/Randalgs11/lectures/lect0209.pdf.

[16] M. Mitzenmacher and E. Upfal, *Probability and Computing: Randomized Algorithms and Probabilistic Analysis*. Cambridge University Press, 2005.

[17] J. L. Carter and M. N. Wegman, "Universal classes of hash functions," *Journal of Computer and System Sciences*, vol. 18, no. 2, pp. 143–154, 1979. DOI: 10.1016/0022-0000(79)90044-8